Explainer: Problem 6 (Rudin 7.14) — The Schoenberg Space-Filling Curve

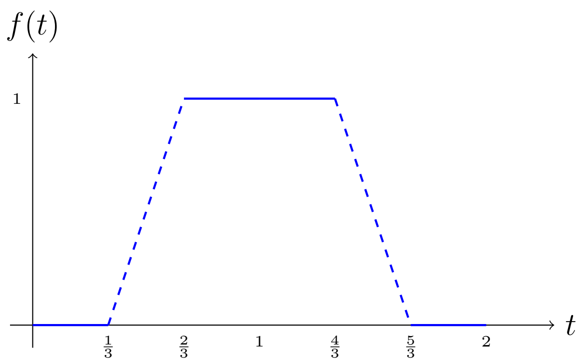
What's going on?

We're constructing a **continuous** curve $\Phi : [0, 1] \to \mathbb{R}^2$ that fills the entire unit square $[0, 1]^2$. This is deeply counterintuitive — a one-dimensional curve covers a two-dimensional region!

The construction uses a helper function f and the Cantor set.

The helper function f

f is periodic with period 2, and on $[0, 1]$:



The key property: f is 0 on $[0, 1/3]$ and 1 on $[2/3, 1]$. It transitions smoothly in between. By periodicity, f reads off the "digits" of a base-3 expansion.

The curve $\Phi(t) = (x(t), y(t))$

$$x(t) = \sum_{n=1}^{\infty} 2^{-n} f(3^{2n-1}t), \qquad y(t) = \sum_{n=1}^{\infty} 2^{-n} f(3^{2n}t).$$

The idea: $3^k t$ "shifts" the base-3 expansion of t by k digits. Applying f reads off whether that digit is 0 or 2 (the Cantor set digits). The 2^{-n} weighting assembles these into a binary expansion.

Continuity of Φ

Each term $2^{-n} f(3^k t)$ is continuous (it's a continuous function composed and scaled). The series converges uniformly by the Weierstrass M-test ($|2^{-n} f(3^k t)| \le 2^{-n}$ and $\sum 2^{-n} < \infty$). By Theorem 7.12, $x(t)$ and $y(t)$ are continuous, hence Φ is continuous.

Surjectivity: why Φ hits every point in $[0,1]^2$

Pick any $(x_0, y_0) \in [0,1]^2$. Write their binary expansions:

$$x_0 = \sum_{n=1}^{\infty} 2^{-n} a_{2n-1}, \qquad y_0 = \sum_{n=1}^{\infty} 2^{-n} a_{2n},$$

where each $a_i \in \{0,1\}$. Now **interleave** these bits into a base-3 number:

$$t_0 = \sum_{i=1}^{\infty} \frac{2a_i}{3^{i+1}}.$$

Target: (x_0, y_0)

x_0: bits $a_1, a_3, a_5, \ldots$ $\qquad$ y_0: bits $a_2, a_4, a_6, \ldots$

$\boxed{t_0 = 0.0(2a_1)(2a_2)(2a_3)(2a_4)\ldots \text{ in base } 3}$

Each digit is 0 or 2 $\Rightarrow$
t_0 is in the **Cantor set**!

Since each $2a_i \in \{0,2\}$, the number t_0 has only digits 0 and 2 in base 3 — it's in the Cantor set. One then checks $f(3^k t_0) = a_k$, so $\Phi(t_0) = (x_0, y_0)$.

Proof outline

1. **Continuity:** Weierstrass M-test with $M_n = 2^{-n}$.
2. **Surjectivity:** Given (x_0, y_0), construct t_0 in the Cantor set by interleaving binary digits. Verify $f(3^k t_0) = a_k$ using f's definition on $[0, 1/3]$ and $[2/3, 1]$.
3. **Cantor set:** t_0 has only digits $0, 2$ in base 3, so $t_0 \in C$.